Temporal Disaggregation and the Fragility of Fiscal Leakage Estimates: Evidence from Timor-Leste

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Abstract

Data-limited economies often disaggregate annual national accounts before estimating quarterly fiscal models. This paper shows that both the disaggregation step and the sample window can shape the central result. Using official quarterly national accounts for Timor-Leste from 2014Q1 to 2023Q4, a Bayesian VAR and local projections find no robust evidence that a government expenditure shock produces a large or persistent increase in imports. The import response is small, between −0.5% and 0.8% across horizons, while the response of non-oil GDP is small and short-lived. In contrast, the same models estimated on a 2002–2023 sample built by Denton–Cholette temporal disaggregation generate a large, precisely estimated import response, peaking near 2.1% in the BVAR and 26% in local projections. A 2x2 hybrid design on the common 2014–2023 window shows that neither interpolating the fiscal series nor interpolating the outcome series reproduces this leakage. The long-sample result is therefore not a mechanical consequence of interpolation alone: it is concentrated in the earlier observations and is not robust in the common window. The results caution against using long interpolated samples as a substitute for official high-frequency data and show why common-window tests are essential.

JEL classification: C32, E62, O53.

Keywords: temporal disaggregation; fiscal leakage; Bayesian VAR; local projections; Timor-Leste.

1 Introduction

Timor-Leste is a small, open, petroleum-rich economy with a short macroeconomic history. Government expenditure finances state-building, infrastructure, and a large share of domestic absorption, while the productive base remains narrow. A central policy question is therefore whether fiscal expansion supports domestic non-oil activity or leaks abroad through imports. Because the country's national accounts are available at annual frequency for most of the sample, earlier versions of this analysis relied on temporally disaggregated quarterly data to estimate dynamic fiscal models.

This paper revisits that question with official quarterly data. The Instituto Nacional de Estatística (INETL) publishes quarterly GDP by expenditure from 2014 onwards. We use these data to estimate a quarterly Bayesian VAR (BVAR) and local projections (LP) for 2014Q1–2023Q4, and

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we compare the results with those obtained from a long 2002–2023 sample constructed by univariate Denton–Cholette disaggregation. The comparison is designed to separate the estimator, the data construction, and the sample window.

The main finding is that the strong leakage result is fragile. In the official sample, a three-variable BVAR with government expenditure, imports, and non-oil GDP is stable and estimates an import response of about -0.03% in the first quarter, 0.09% after three quarters, and essentially zero thereafter; the 68% interval includes zero at all horizons. Local projections confirm that the official import response is small, between -0.5% and 0.8% depending on the horizon, and economically negligible relative to the long-sample estimates. Non-oil GDP shows a small impact response in the BVAR but no robust positive response in the LP. The five-variable official BVAR, which adds the CPI and private credit, is unstable because the short sample cannot support the additional dynamics; this instability is itself informative about the limits of the official sample.

In the long interpolated sample, the same three-variable BVAR is stable and estimates a significant import response that rises to about 2.1% after six to seven quarters. The LP estimates are larger, peaking near 26% after two to three quarters. The long-sample results reproduce the earlier finding: imports respond earlier and more strongly than non-oil GDP. The contrast between the two samples is stark, and it survives alternative orderings and seasonal adjustment choices. It does not, however, survive a common-window test: in a 2x2 design that varies the fiscal series and the outcome series separately over 2014–2023, no cell reproduces the leakage. Interpolation alone is therefore not sufficient to generate the long-sample result.

The contribution is threefold. First, the paper provides BVAR and LP estimates for Timor-Leste based on official quarterly national accounts, rather than interpolated annual data. Second, it documents the fragility of the earlier leakage result and develops a simple 2x2 decomposition that separates fiscal interpolation, outcome interpolation, and the sample window. This decomposition shows that the earlier mechanism was too broad: long interpolated samples can produce estimates that are not reproduced in a common official window, but interpolation by itself does not mechanically generate them. Third, it offers a practical warning for policy analysis in data-limited economies: estimates of fiscal leakage based on long interpolated samples should be treated with caution, and they should be tested against official high-frequency data and common-window designs whenever available.

The remainder of the paper is organized as follows. Section 2 reviews the literature. Section 3 describes the data and the construction of the long sample. Section 4 presents the empirical strategy. Section 5 reports the results and robustness checks. Section 6 concludes.

2 Literature

The macroeconomic effects of fiscal policy have been extensively studied, but consensus on the size of fiscal multipliers remains elusive, especially in developing and resource-rich economies. Blanchard and Perotti (2002) introduced the structural VAR approach to identify government spending shocks and showed that spending increases typically raise output. Mountford and Uhlig (2009) used sign restrictions to show that fiscal effects depend on the type of shock and the state of the economy. For small open economies, the literature emphasizes that multipliers tend to be smaller because part of the stimulus is absorbed by imports. Ilzetzki et al. (2013) find lower multipliers in economies with more open trade regimes and more flexible exchange rates. Evidence for Southeast Asia, including ASEAN member states, points in the same direction: fiscal multipliers are typically modest and sensitive to the import content of spending (Tang et al. 2013; Dime et al. 2021); a calibration for Cambodia also implies a low first-year multiplier (Baksa et al. 2020). Resource-rich economies face additional challenges: commodity-price volatility and Dutch-disease pressures complicate the conduct of fiscal policy (Frankel 2012).

Vector autoregressions are the standard tool for dynamic multivariate analysis, but short time series make unrestricted OLS estimates unreliable. Bayesian VARs, introduced to macroeconomics by Litterman (1986) and refined by Giannone et al. (2015), impose prior beliefs that shrink the parameter space and can produce stable responses even with limited observations. The `BVAR` package of Kuschnig and Vashold (2021) implements hierarchical prior selection for these models.

A separate literature addresses the construction of high-frequency data from annual benchmarks. Denton (1971) proposed a quadratic minimization method, extended by Cholette (1984); Sax and Steiner (2013) and the `tempdisagg` package implement these methods in R. Temporal disaggregation is widely used in data-poor settings, but it is not a source of new information: it distributes an annual total using a smoothness criterion or a related indicator. The resulting quarterly series are mechanically smooth and share the annual benchmark with the other disaggregated variables. This paper asks whether such shared smoothness is sufficient to generate spurious fiscal transmission. The answer is negative: shared smoothness can distort estimates, but its effect depends on the sample window, so it cannot be inferred from a long interpolated sample alone.

Studies of fiscal policy in Timor-Leste are scarce, but the available estimates point to small multipliers. World Bank (2021) estimates an impact multiplier of 0.17 and a cumulative multiplier of 0.08 using a quarterly SVAR for 2010–2019 with real quarterly data. International Monetary Fund (2024) reports multipliers of 0.1–0.2, attributing the small size to the low quality and high import content of public spending. These estimates are consistent with the small official-data responses documented below, and they are difficult to reconcile with the large leakage implied by the long interpolated sample. This paper contributes by combining official quarterly data with a transparent comparison to the interpolated sample, and by showing that the long-sample result is fragile in a common-window design.

3 Data

3.1 Official quarterly data

The baseline data are the official quarterly national accounts published by INETL for 2004–2023 (Instituto Nacional de Estatística 2024). We use the expenditure table (Table 7.1) at constant 2015 prices for 2014Q1–2023Q4, the period for which quarterly observations are available. The sample contains 40 quarters. The variables are general government final consumption expenditure, non-oil GDP by expenditure, and imports of goods and services. The consumer price index is obtained from the monthly DNE series and averaged to quarterly frequency (Direcção Nacional de Estatística 2025). Credit to the private sector is the BCTL monetary survey series, measured at the end of each quarter (Banco Central de Timor-Leste 2024).

The official series are seasonally adjusted with X-13ARIMA-SEATS before estimation. The baseline uses the seasonally adjusted data; an unadjusted specification with quarterly dummies is reported as a robustness check. Table 1 summarizes the variables and sources.

3.2 Data validation and source differences

The official quarterly GDP and import series aggregate almost exactly to the annual ADB series (Asian Development Bank 2025). Over 2014–2023, the largest absolute difference between the sum of the four quarterly observations and the annual figure is 0.85 million USD for GDP (2016) and 1.10 million USD for imports (2021). These differences are negligible relative to annual levels of about 1,500–1,650 million USD for GDP and 730–980 million USD for imports.

Table 1: Variables, sources, and samples

Variable	Interpretation	Official baseline (2014–2023)	Long-sample comparison (2002–2023)
`ln_gov_exp`	Government expenditure	General Government FCE, INETL	ADB government expenditure, interpolated
`ln_gdp_non`	Real non-oil GDP	GDP by expenditure, INETL	ADB non-oil GDP, interpolated
`ln_imports`	Real imports	Imports of goods and services, INETL	ADB imports, interpolated
`ln_cpi`	Consumer price index	DNE monthly CPI, quarterly average	DNE/ADB CPI
`ln_credit`	Private credit	BCTL domestic credit, end-of-quarter	ADB private credit

Note: All real variables are in millions of USD at constant 2015 prices. The official sample is 2014Q1–2023Q4 (40 observations); the long sample is 2002Q1–2023Q4 (88 observations). The baseline BVAR uses government expenditure, imports, and non-oil GDP; the CPI and credit enter the five-variable robustness models.

Two source differences require attention. First, official General Government FCE is systematically lower than the ADB government expenditure series. The gap ranges from 180.5 million USD in 2014 to 123.9 million USD in 2023 and matches the on-budget Development and Technical Assistance component that the ADB reports separately. The ADB broad aggregate is therefore the sum of the narrow national-accounts measure and this component; Table 3 reports the reconciliation. This matters for the interpretation of the hybrid design: the broader fiscal measure is not an arbitrary alternative, but the ADB aggregate that earlier work used. Second, the BCTL domestic credit series matches the ADB private credit series until 2017, but is about 50 million USD lower from 2018 to 2022 and 6 million USD lower in 2023. The break is consistent with a definitional change in the monetary survey. We use the BCTL series for the official sample and report the discrepancy rather than forcing the two sources to match. Table 2 summarizes the comparison.

Table 2: Comparison of official quarterly and annual ADB series, 2014–2023

Series	Comparison	Result
GDP	Sum of quarters vs annual ADB	Max. absolute difference 0.85 (2016)
Imports	Sum of quarters vs annual ADB	Max. absolute difference 1.10 (2021)
Government FCE	Sum of quarters vs ADB government expenditure	Gap equals Development & TA (on-budget)
Private credit	Q4 vs annual ADB	About 50 million USD lower in 2018–2022

Note: Differences are in millions of USD. Source: Author's elaboration from INETL, ADB, and BCTL data.

Table 3: Reconciliation of government expenditure, 2014–2023 (millions of USD)

Year	Official GFC	Development & TA	ADB broad aggregate
2014	724	180	904
2015	776	161	937
2016	773	154	927
2017	718	155	873
2018	726	144	870
2019	769	129	898
2020	816	126	942
2021	834	135	969
2022	833	133	966
2023	834	124	958

Note: Official GFC is General Government final consumption expenditure from INETL Table 7.1. Development & TA is the on-budget Development and Technical Assistance component that reconciles the national-accounts measure with the ADB broad government expenditure series. Values are rounded. Source: Author's elaboration from INETL and ADB data; see `Data/csv/gov_fce_decomposition.csv`.

3.3 The long interpolated sample

The long sample covers 2002Q1–2023Q4 (88 quarters) and is the sample used in earlier versions of this project. Annual series from the ADB Key Indicators (Asian Development Bank 2025) are disaggregated with the univariate Denton–Cholette proportional method (Denton 1971; Cholette 1984; Sax and Steiner 2013; Sax and Steiner 2025). For flows, the annual sum constraint is imposed; for the CPI and credit, the quarterly average constraint is imposed. The quarterly import indicator from Trade Map (International Trade Centre 2025) was tested in a Chow–Lin specification but was not statistically significant (coefficient 0.171, standard error 0.296, $p = 0.571$), so the univariate method was retained. The long sample includes five event dummies: the 2006 political crisis, the 2009 global financial crisis, the 2012–2014 transition, the 2016–2018 political crisis, and the COVID-19 pandemic.

The key feature of the long sample is that all real series are smooth by construction. Table 4 compares the standard deviation of quarterly log changes over the common 2014–2023 period. For government expenditure, the official series is about 15 times more volatile than the interpolated series; for GDP, about 14 times; for imports, about 4 times. The official data contain high-frequency variation that the interpolation removes. This difference is central to the interpretation of the results.

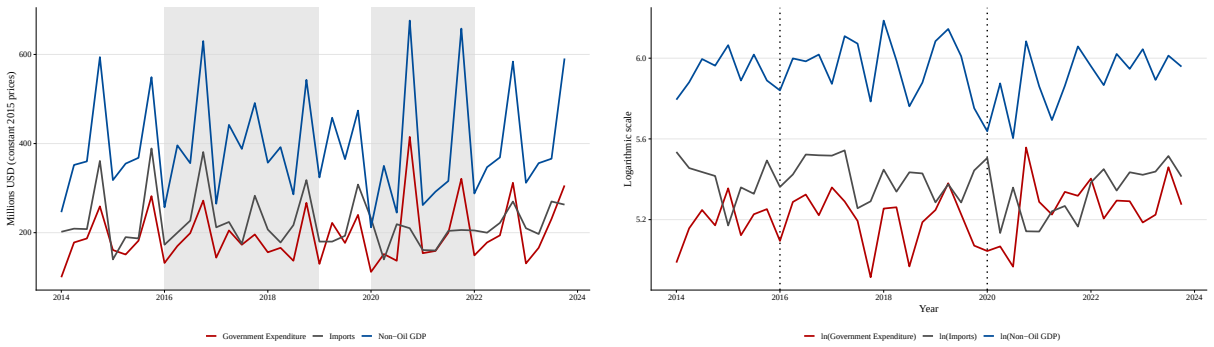
Table 4: Volatility of quarterly log changes, 2014–2023

Series	Official quarterly data	Long interpolated sample
Government expenditure	0.181	0.012
Non-oil GDP	0.185	0.013
Imports	0.136	0.033

Note: Standard deviation of the first difference of the log series. The official values use the seasonally adjusted series; the long-sample values are computed over 2014–2023, the period common to both samples. Source: Author's elaboration.

3.4 Descriptive evidence

Figure 1 shows the official series in levels and logs. Government expenditure is volatile and falls sharply in the first half of the sample, then recovers around the 2016–2018 period and the pandemic. Non-oil GDP grows steadily, with a dip in 2020. Imports fluctuate with construction and pandemic-related demand. The shaded areas mark the 2016–2018 and 2020–2021 event windows.



(a) Levels **(b)** Logarithms

Figure 1: Official quarterly macroeconomic series, 2014Q1–2023Q4. Gray areas mark the 2016–2018 and 2020–2021 event windows. Units: millions of USD at constant 2015 prices. Source: Author's elaboration from INETL, DNE, and BCTL data.

Figure 6 in Appendix A shows the corresponding long-sample series. The contrast in high-frequency variation is visible: the interpolated series follow smooth trends, while the official series display substantial quarter-to-quarter variation.

4 Empirical strategy

4.1 Bayesian VAR

The baseline official model is a three-variable BVAR in logarithms:

$$y_t = (\ln \text{gov}_t,\ \ln \text{imports}_t,\ \ln \text{GDP}_t)',$$

with two lags, a constant, and the two event dummies. The variables are ordered with government expenditure first, so the fiscal shock is recursively identified: government expenditure does not respond contemporaneously to imports or GDP, while both can respond to the fiscal shock within the quarter. A second ordering places GDP first and government expenditure second, allowing expenditure to react to activity within the quarter. The comparison between orderings is a sensitivity check, not a test that eliminates all identification assumptions.

The prior is a Minnesota prior with hierarchical selection of the overall tightness λ. The prior mode of λ is 0.2, with support $[0.001, 3]$; α has mode 2 and support $[1, 3]$; the scale ψ is set to the square root of the variance of the first differences of each series, with the package's default bounds. The sampler uses 100,000 draws, 20,000 burn-in draws, and thinning every 20 draws, leaving 4,000 retained draws. Impulse responses are computed for 12 quarters with 68% and 90% credible intervals.

The three-variable specification is chosen because the official sample has only 40 observations. A five-variable model that adds the CPI and private credit has a maximum companion modulus above one (Table 5); the additional dynamics are not supported by the data. The five-variable model is reported as a robustness check, not used for inference. The long sample, with 88 observations, supports both the three- and five-variable specifications.

4.2 Local projections

Local projections provide a complementary estimate that does not rely on the full VAR dynamics (Jordà 2005). We estimate a three-variable VAR with the same variables, lags, and dummies, extract the recursively identified government expenditure shock, and project each outcome on the shock and its pre-determined controls:

$$y_{t+h} = \alpha_h + \beta_h \varepsilon_t + \Gamma_h X_{t-1} + u_{t+h},$$

where X_{t-1} contains a constant, two lags of the three core variables, and the event dummies. The coefficient β_h is the response at horizon h. For government expenditure, imports, and non-oil GDP we implement the IV local-projection estimator of Adämmer (2019) using `lpirfs`; for the CPI and private credit we use the same estimator with the two core lags as exogenous controls, so that all five responses share the same shock and information set. The HAC bandwidth is fixed at six quarters, and standard errors are finite-sample adjusted. The shock is normalized so that the impact response of government expenditure is 1%, which makes the LP coefficients directly comparable to the BVAR elasticities. We report horizons 0–12.

4.3 Stability and stationarity

Table 5 reports the maximum modulus of the companion matrix for each specification. The official three-variable model is stable (0.795), as are the long-sample three- and five-variable models

(0.983 and 0.993). The official five-variable model is unstable (1.089), and we do not interpret its impulse responses. All four 2x2 cells are stable over the common 2014–2023 window. The pre-2014 subsample alone is not stable (1.015), so we report its responses only as a descriptive diagnostic and do not use them for inference.

Table 5: Companion-matrix stability

Model	Sample	Observations	Max. modulus	Stable
3-variable BVAR	Official, 2014–2023	40	0.795	Yes
5-variable BVAR	Official, 2014–2023	40	1.089	No
3-variable BVAR, NSA	Official, 2014–2023	40	0.701	Yes
3-variable BVAR	Long, 2002–2023	88	0.983	Yes
5-variable BVAR	Long, 2002–2023	88	0.993	Yes
3-variable BVAR, Hybrid-F	2014–2023	40	0.940	Yes
3-variable BVAR, Hybrid-O	2014–2023	40	0.933	Yes
3-variable BVAR, Interpolated both	2014–2023	40	0.984	Yes
3-variable BVAR, Interpolated both	2002–2013	48	1.015	No

Note: Maximum modulus of the eigenvalues of the companion matrix at the posterior mean. A value below one indicates a stable VAR. Source: Author's elaboration.

Unit-root tests are reported in Table 6. We apply augmented Dickey–Fuller (Dickey and Fuller 1979), KPSS (Kwiatkowski et al. 1992), and Zivot–Andrews (Zivot and Andrews 1992) tests. The evidence is mixed: ADF tests do not reject a unit root for most series at the 5% level, while KPSS tests reject stationarity for the CPI and credit. The Zivot–Andrews test rejects the unit root for non-oil GDP with a break in 2019Q4, but not for the other series. Given the short sample and the mixed evidence, we estimate the BVAR in levels with a Minnesota prior, which shrinks toward a random-walk representation and is robust to persistent series. The long-sample tests are reported in Appendix A.

Table 6: Unit-root and stationarity tests, official sample

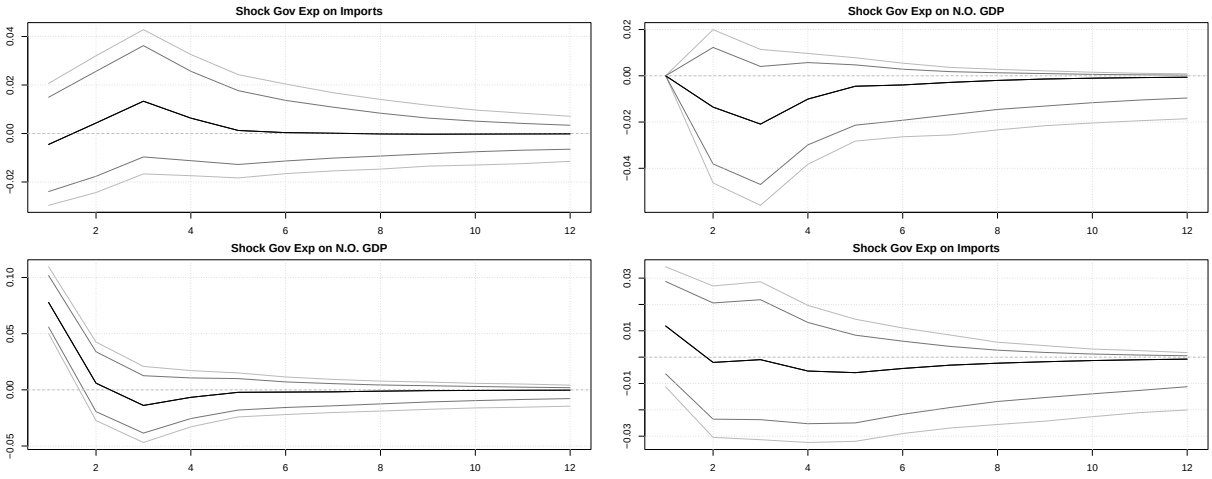
Series	ADF (p)	KPSS (p)	ZA (t)	5% CV	Break	Stationary?
`ln_gov_exp`	0.093	0.100	−4.66	−5.08	2020Q3	No
`ln_gdp_non`	0.259	0.100	−6.24	−5.08	2019Q4	Yes
`ln_imports`	0.360	0.100	−3.85	−5.08	2020Q1	No
`ln_cpi`	0.976	0.010	−2.76	−5.08	2019Q4	No
`ln_credit`	0.764	0.010	−3.77	−5.08	2021Q1	No

Note: ADF: H_0 unit root. KPSS: H_0 stationarity. ZA: H_0 unit root with a structural break (model "both", two lags); a t-statistic below the critical value rejects H_0. Source: Author's elaboration.

5 Results

5.1 Official BVAR

Figure 2 shows the impulse responses to a one-unit structural government expenditure innovation under the two orderings. The responses are rescaled to a 1% increase in government expenditure in Table 7; the figures preserve the original shock scale. The fiscal shock is persistent but transitory: government expenditure itself rises and then returns toward zero. Imports do not display the large positive response found in the long sample. Under order 1, the median import response is −0.03% at impact, 0.09% after three quarters, and essentially zero thereafter; the 68% interval includes zero at all horizons. The GDP response is positive at impact (0.52%) but reverses by the third quarter (−0.09%) and is not distinguishable from zero at longer horizons.



(a) Order 1 **(b)** Order 2

Figure 2: Official-sample BVAR impulse responses to a government expenditure shock, orders 1 and 2. Bands: 68% and 90%. Units: response to a one-unit structural innovation. Table 7 reports the responses rescaled to a 1% increase in government expenditure. Source: Author's elaboration.

Table 7: BVAR responses to a 1% government expenditure shock (percent)

Response	Horizon	Official 3-variable (2014–2023)	Long 3-variable (2002–2023)	Long 5-variable (2002–2023)
Imports	1	−0.03 [−0.16, 0.10]	0.34 [0.21, 0.48]	0.35 [0.22, 0.49]
Imports	2	0.03 [−0.12, 0.17]	0.78 [0.55, 1.02]	0.81 [0.59, 1.05]
Imports	3	0.09 [−0.06, 0.24]	1.22 [0.87, 1.57]	1.25 [0.92, 1.58]
Imports	6	0.00 [−0.08, 0.09]	2.08 [1.48, 2.73]	2.00 [1.45, 2.62]
Imports	7	0.00 [−0.07, 0.07]	2.15 [1.49, 2.91]	2.05 [1.44, 2.75]
Imports	12	0.00 [−0.04, 0.02]	1.30 [0.50, 2.33]	1.26 [0.41, 2.31]
Non-oil GDP	1	0.52 [0.37, 0.68]	−0.01 [−0.06, 0.04]	−0.01 [−0.06, 0.04]
Non-oil GDP	2	0.04 [−0.13, 0.23]	0.02 [−0.07, 0.11]	0.00 [−0.08, 0.07]
Non-oil GDP	3	−0.09 [−0.26, 0.08]	0.08 [−0.05, 0.20]	0.04 [−0.06, 0.14]
Non-oil GDP	6	−0.01 [−0.10, 0.05]	0.31 [0.09, 0.53]	0.29 [0.15, 0.44]
Non-oil GDP	7	−0.01 [−0.09, 0.04]	0.38 [0.13, 0.64]	0.37 [0.22, 0.54]
Non-oil GDP	12	0.00 [−0.05, 0.01]	0.52 [0.17, 0.92]	0.59 [0.36, 0.86]

Note: Posterior medians with 68% credible intervals in brackets. Responses are rescaled so that the impact response of government expenditure is 1%. A value of 0.34 means that a 1% increase in government expenditure is associated with a 0.34% increase in imports. The long 5-variable model is the specification used in earlier versions of the paper. Source: Author's elaboration.

The five-variable official BVAR produces qualitatively similar results but is unstable. Its import response is small and the GDP response is short-lived. Because the maximum companion modulus is 1.089, we do not use it for inference. The instability is a symptom of the short sample: with 40 observations and two lags, the five-variable system has more parameters than the data can support. This is not a defect of the estimator; it is a constraint of the available official data.

5.2 Official local projections

Figure 3 and Table 8 report the local-projection estimates. The official import response is small and imprecisely estimated: 0.52% at impact, 0.10% after one quarter, −0.42% after two quarters, 0.16% after three quarters, and 0.48–0.51% after six to twelve quarters. The negative response at two quarters is statistically distinguishable from zero at the 68% level, but its magnitude is negligible and it is offset at the other horizons. All import responses lie between −0.5% and 0.8%. Non-oil GDP is close to zero after the impact quarter, with a negative point estimate at impact (−0.37%), a positive but insignificant estimate after one quarter (0.23%), and a negative estimate at six quarters (−0.37%). The CPI and credit responses are close to zero.

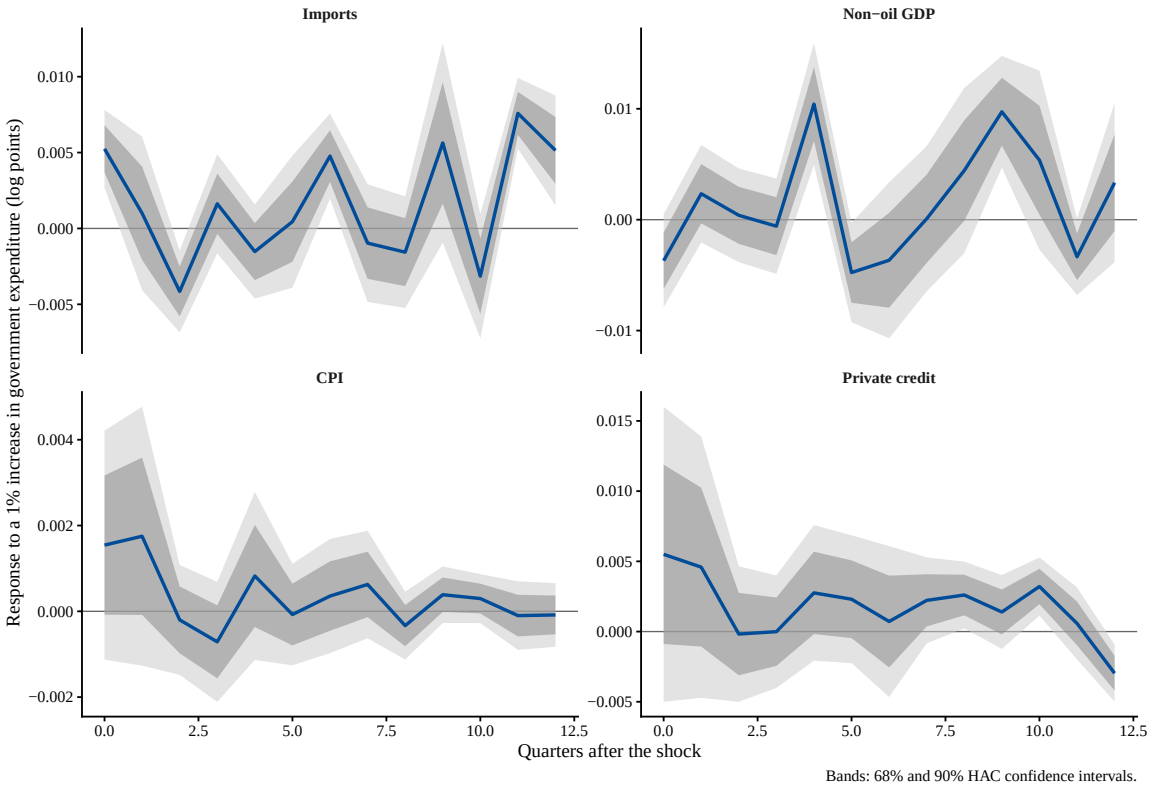


Figure 3: Official-sample local projections to a 1% government expenditure shock. Bands: 68% and 90% Newey–West HAC confidence intervals. Source: Author's elaboration.

Taken together, the official BVAR and LP provide no evidence of a large or persistent import leakage. The BVAR estimates an import response below 0.1% at short horizons; the LP estimates a small positive response of about 0.5% at longer horizons, but it is an order of magnitude smaller than the long-sample estimates and does not support the view that fiscal stimulus is largely spent abroad. The GDP response is fragile: the BVAR finds a short-lived positive impact effect, while the LP finds a small negative effect at impact and six quarters. The robust conclusion is that the official data do not identify a positive output effect that persists.

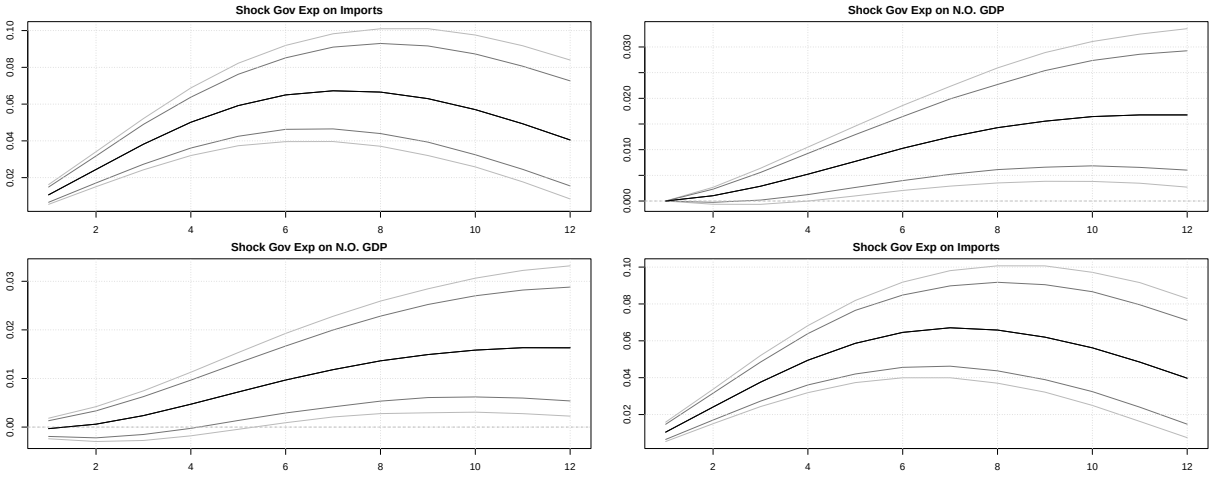
Table 8: Local-projection responses to a 1% government expenditure shock (percent)

Response	Horizon	Official sample	Long sample
Imports	0	0.52 [0.37, 0.68]	9.06 [7.99, 10.13]
Imports	1	0.10 [−0.21, 0.41]	19.52 [17.24, 21.80]
Imports	2	−0.42 [−0.58, −0.25]	26.14 [22.68, 29.60]
Imports	3	0.16 [−0.04, 0.36]	25.97 [21.02, 30.92]
Imports	6	0.48 [0.31, 0.65]	0.31 [−9.05, 9.66]
Imports	12	0.51 [0.29, 0.73]	11.82 [3.76, 19.89]
Non-oil GDP	0	−0.37 [−0.62, −0.12]	0.05 [−0.48, 0.57]
Non-oil GDP	1	0.23 [−0.03, 0.50]	0.35 [−0.76, 1.47]
Non-oil GDP	2	0.04 [−0.22, 0.30]	0.84 [−0.70, 2.38]
Non-oil GDP	3	−0.06 [−0.32, 0.20]	1.42 [−0.29, 3.14]
Non-oil GDP	6	−0.37 [−0.79, 0.06]	2.81 [0.43, 5.20]
Non-oil GDP	12	0.33 [−0.10, 0.77]	6.78 [4.46, 9.10]

Note: Point estimates with 68% Newey–West HAC confidence intervals in brackets. The shock is normalized so that the impact response of government expenditure is 1%. Source: Author's elaboration.

5.3 The long-sample result

For comparison, Figure 4 and the last two columns of Table 7 report the long-sample BVAR. The same three-variable specification is stable and estimates a significant import response: 0.34% at impact, 1.22% after three quarters, 2.08% after six quarters, and 2.15% at the peak after seven quarters. The 68% interval excludes zero at every horizon shown. Non-oil GDP responds with a lag and remains modest: 0.31% after six quarters and 0.52% after twelve. The five-variable long-sample model, which is the specification used in earlier work, gives nearly identical results (0.35% at impact, 2.05% at the peak).



(a) Order 1 **(b)** Order 2

Figure 4: Long-sample BVAR impulse responses to a government expenditure shock, orders 1 and 2. Bands: 68% and 90%. Units: response to a one-unit structural innovation. Table 7 reports the responses rescaled to a 1% increase in government expenditure. Source: Author's elaboration.

The long-sample LP estimates are larger still (Figure 5 and Table 8). The import response rises from 9.1% at impact to 19.5% after one quarter, 26.1% after two quarters, and 26.0% after three quarters, then becomes imprecise at intermediate horizons before returning to 11.8% after twelve quarters. The GDP response rises gradually to 6.78% after twelve quarters. The LP and BVAR agree on the qualitative pattern: in the long sample, imports respond earlier and more strongly than GDP. The magnitude differs because the LP imposes less structure and estimates

the response separately at each horizon; the LP estimates are larger at the peak but less stable across horizons, and they should be read as complementary evidence rather than a strict upper bound.

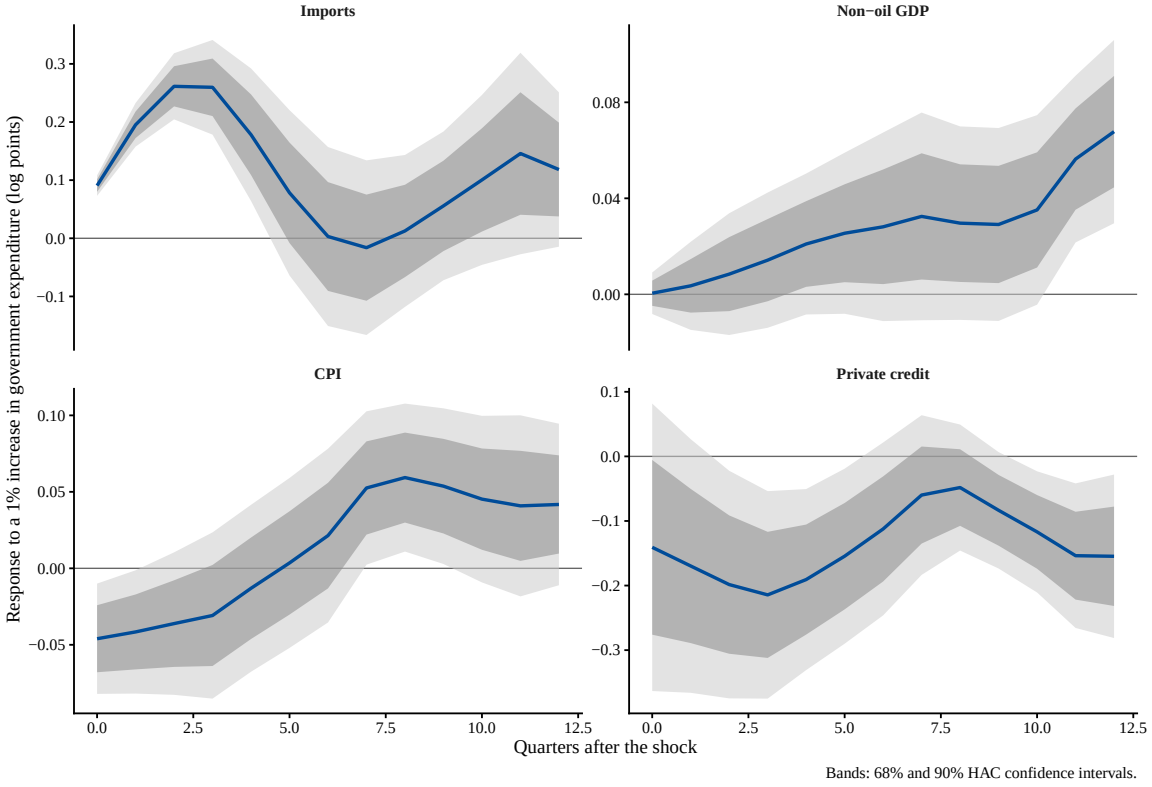


Figure 5: Long-sample local projections to a 1% government expenditure shock. Bands: 68% and 90% Newey–West HAC confidence intervals. Source: Author's elaboration.

5.4 Why the two samples disagree

The contrast between the official and long-sample results is not driven by the estimator, the prior, or the identification. The same three-variable BVAR, with the same lags, prior, and ordering, produces a near-zero import response in the official sample and a large, significant response in the long sample. The difference must therefore come from the data: either their construction or the sample period.

A simple 2x2 design separates the two dimensions. We estimate four cells over the common 2014–2023 window: the official baseline; Hybrid-F, which replaces official government expenditure with the interpolated ADB broad series while keeping official outcomes; Hybrid-O, which keeps official government expenditure and replaces the outcomes with interpolated series; and Interpolated both, which uses interpolated fiscal and outcome series. All cells use the same three-variable BVAR, two lags, prior, sampler, event dummies, and ordering. Table 9 reports the import response at horizons 1, 3, 6, and 12. No cell reproduces the large and persistent leakage. Hybrid-F has wide intervals and an import response between −0.28% and 0.18%. Hybrid-O is close to zero at every horizon. Interpolated both is −0.04% after one quarter and 0.31% after twelve quarters, with intervals that include zero throughout. This is a direct test of the interpolation hypothesis: if joint interpolation were sufficient to manufacture the result, the Interpolated both cell should reproduce it over the common window. It does not.

The long sample differs from the common-window interpolated cell because it also includes 2002–2013. Table 10 reports a descriptive decomposition. On 2002–2013, the point estimates

Table 9: Common-window 2x2 hybrid design, 2014–2023 (percent)

Cell	Fiscal series	Outcome series	Imports $h = 1$	Imports $h = 3$	Imports $h = 6$	Imports $h = 12$
Official baseline	Official GFC	Official	−0.03 [−0.16, 0.10]	0.09 [−0.06, 0.24]	0.00 [−0.08, 0.09]	0.00 [−0.04, 0.02]
Hybrid-F	Interpolated ADB broad	Official	−0.28 [−1.31, 0.76]	−0.03 [−0.80, 0.82]	0.18 [−0.84, 1.25]	0.16 [−0.86, 1.29]
Hybrid-O	Official GFC	Interpolated	0.00 [−0.04, 0.04]	0.01 [−0.06, 0.08]	0.03 [−0.05, 0.11]	0.02 [−0.04, 0.12]
Interpolated both	Interpolated ADB broad	Interpolated	−0.04 [−0.38, 0.31]	0.05 [−0.32, 0.42]	0.16 [−0.30, 0.69]	0.31 [−0.35, 1.21]

Note: Posterior medians with 68% credible intervals in brackets. Responses are rescaled to a 1% increase in government expenditure and use order 1. All four cells are stable; see Table 5. The official baseline is the same model as in Table 7. Source: Author's elaboration.

are large (0.34% after one quarter, 1.27% after three, 1.97% after six, and 1.30% after twelve), but the maximum companion modulus is 1.015, so that specification is unstable and we do not interpret it. On 2014–2023, the interpolated cell is close to zero, while the full 2002–2023 model is stable and reproduces the leakage. The gap is therefore associated with the earlier observations, not with interpolation in the common window. Because the pre-2014 subsample is unstable, we cannot separately identify whether the earlier result reflects period-specific shocks, an interaction between those shocks and interpolation, or the particular annual variation in that period. We treat it as a cautionary finding rather than a structural estimate.

Table 10: Period decomposition of the import response, order 1 (percent)

Sample	Observations	Max. modulus	Imports $h = 1$	Imports $h = 6$	Imports $h = 12$
Official, 2014–2023	40	0.795	−0.03	0.00	0.00
Interpolated, 2014–2023	40	0.984	−0.04	0.16	0.31
Interpolated, 2002–2013	48	1.015	0.34	1.97	1.30
Interpolated, 2002–2023	88	0.983	0.34	2.08	1.30

Note: Median import responses to a 1% government expenditure shock. The 2002–2013 model is unstable and is reported for diagnosis only. The 2014–2023 official and interpolated rows are the corresponding cells of Table 9. Source: Author's elaboration.

This reinterpretation qualifies the earlier mechanism. Denton–Cholette does smooth the series and removes high-frequency independent variation (Table 4), but shared smoothness alone is not enough to produce a large import response in 2014–2023. Conversely, the official data contain independent quarterly variation, yet the absence of leakage is not simply a mechanical consequence of that variation either. The long-sample result is a joint product of the earlier sample and the construction of the data, and the common-window design shows why it should not be generalized.

A final check addresses seasonal adjustment. The unadjusted official model, which uses quarterly dummies instead of X-13 adjustment, produces a short-lived first-quarter import spike (0.39%) but no persistent response; the spike disappears by the second quarter. Seasonal adjustment is therefore not driving the official result. The broader fiscal measure alone does not restore the leakage (Hybrid-F), interpolated outcomes alone do not restore it (Hybrid-O), and joint interpolation in the common window does not restore it (Interpolated both).

The five-variable official model is unstable, but its point responses are also small and short-lived. We do not interpret them because the companion roots exceed one. The instability is consistent with the short sample: adding the CPI and credit doubles the number of parameters without adding enough observations. The long sample supports the five-variable model, which is why earlier work could estimate it; the cost is that the additional observations are interpolated and come from an earlier period.

5.5 Robustness

Table 11 collects the alternative specifications. The official unadjusted model is stable and produces a first-quarter import response of 0.39%, but the response turns negative in the second and third quarters and is essentially zero thereafter. The official five-variable model is unstable and is not used for inference. Hybrid-F is stable and produces import responses between −0.3% and 0.2%, with intervals that include zero. The long three- and five-variable models are stable and produce significant import responses of about 2% at the peak. The qualitative conclusion is unchanged across specifications: the official data do not support a large, persistent leakage; the long interpolated data do.

Table 11: Robustness: stability and selected responses

Model	Sample	Max. modulus	Imports $h = 1$	Imports $h = 3$	Imports $h = 12$	GDP $h = 3$
Official 3-var	2014–2023	0.795	−0.03	0.09	0.00	−0.09
Official 5-var	2014–2023	1.089	−0.06	0.05	0.00	−0.05
Official 3-var, NSA	2014–2023	0.701	0.39	−0.09	0.00	0.04
Long 3-var	2002–2023	0.983	0.34	1.22	1.30	0.08
Long 5-var	2002–2023	0.993	0.35	1.25	1.26	0.04
Hybrid-F 3-var	2014–2023	0.940	−0.28	−0.03	0.16	−0.19
Official 3-var, first differences	2014–2023	0.586	−0.06	−0.07	−0.08	0.49

Note: Responses are medians in percent, rescaled to a 1% increase in government expenditure. Hybrid-F uses the interpolated ADB government series with official GDP and imports. The first-difference row reports cumulative responses. The official 5-variable model is unstable and its responses are reported only for completeness. Source: Author's elaboration.

The first-difference specification provides an additional check on the levels model. Estimating the official three-variable system in log differences with the same prior and two lags yields a stable companion matrix (maximum modulus 0.586). Cumulative import responses remain small and slightly negative: −0.06% after one quarter, −0.07% after six quarters, and −0.08% after twelve quarters, with intervals that include zero. The cumulative GDP response is positive and larger than in the levels model (0.49% after one quarter and 0.22% after twelve quarters), which shows that the output response is specification-sensitive. The import conclusion is unchanged: differencing does not restore the long-sample leakage.

The FEVD tells the same story. In the official three-variable model, the fiscal shock explains 1.2% of import variance at the one-quarter horizon, 6.2% after six quarters, and 6.8% after twelve quarters; for GDP, the shares are 27.5%, 24.5%, and 24.1%. In the long three-variable model, the fiscal shock explains 7.3% of import variance at one quarter, 34.5% after six quarters, and 50.3% after twelve quarters; for GDP, the shares are 0.6%, 4.6%, and 14.9%. The long-sample fiscal shock is a dominant source of import fluctuations; the official fiscal shock is not. Figure 8 presents the FEVDs.

6 Conclusion

This paper revisited the evidence on fiscal leakage in Timor-Leste using official quarterly national accounts. The main conclusion is that the strong leakage result obtained from long interpolated data is fragile. With official data for 2014Q1–2023Q4, a stable three-variable BVAR and local projections estimate a small, imprecise import response, between −0.5% and 0.8% across horizons, and an output response that is not robust across specifications. The long 2002–2023 sample built by Denton–Cholette disaggregation produces a large and significant import response, peaking near 2.1% in the BVAR and 26% in local projections. The difference is not due to the estimator, the prior, or the identification.

The methodological lesson is more specific than the earlier version of this paper suggested. Tem-

poral disaggregation does smooth the series and removes high-frequency independent variation, but shared smoothness alone does not generate the leakage: in the common 2014–2023 window, even the cell with interpolated fiscal and outcome series produces a near-zero import response. The long-sample result is associated with the earlier observations, although the pre-2014 subsample alone is too unstable to identify the separate contribution of period-specific shocks and interpolation. The 2x2 design therefore shows that the earlier mechanism was too broad, while confirming that long interpolated samples can produce estimates that are not reproduced in a common official window.

These findings have implications for policy analysis in data-limited economies. Estimates of fiscal multipliers and leakage based on long interpolated samples should be treated with caution, and they should be tested against official high-frequency data and common-window designs whenever available. For Timor-Leste, the official data do not provide evidence of a large import leakage, but they also do not provide a precise estimate of the fiscal multiplier. The small official-data import responses are consistent with the existing estimates of 0.08–0.17 for the multiplier (World Bank 2021; International Monetary Fund 2024). The short sample, the volatility of government expenditure, and the source differences between national-accounts and administrative data all limit precision. The appropriate conclusion is not that leakage is absent, but that the available official data do not support the large and persistent leakage estimated from the long interpolated sample.

Further work should extend the official sample as new quarterly observations become available, explore alternative fiscal shocks based on budget execution data, and apply the common-window 2x2 design to other small open economies with both annual and official quarterly data.

Data and code availability

We welcome readers to use, replicate, review, and extend this work. All code, processed data, and generated results are available in the public repository https://github.com/Emerson-nic/Timor-Leste-fiscal-bvar. The `README` documents the project structure, data sources, script execution order, and document compilation with LuaLaTeX and Biber. The official data are built by `scripts/00_build_quarterly_official.R`; the BVAR is estimated by `scripts/07_irf_fevd.R` with `SAMPLE=official_3var` or `SAMPLE=long_3var`; the local projections are estimated by `scripts/08_local_projections.R`; the 2x2 hybrid and period decomposition are in `scripts/09_hybrid_robustness.R`; and the first-difference BVAR is in `scripts/10_first_difference_bvar.R`. The dashboard data are prepared by `scripts/build_dashboard_data.R`, and the interactive dashboard is available at https://connect.posit.cloud/em3r/content/01a068c1-99da-638d-1ba0-1254293e5a61.

References

Adämmer, Philipp (2019). "lpirfs: An R Package to Estimate Impulse Response Functions by Local Projections". In: *The R Journal* 11.2, pp. 281–299. DOI: 10.32614/RJ-2019-052.

Asian Development Bank (2025). *Key Indicators Database: Timor-Leste*. Online database. Source file: `tim-key-indicators-2025.xlsx`; accessed 20 October 2026. URL: https://kidb.adb.org/economies/timor-leste.

Bai, Jushan and Pierre Perron (1998). "Estimating and Testing Linear Models with Multiple Structural Changes". In: *Econometrica* 66.1, pp. 47–78. DOI: 10.2307/2998540.

— (2003). "Computation and Analysis of Multiple Structural Change Models". In: *Journal of Applied Econometrics* 18.1, pp. 1–22. DOI: 10.1002/jae.659.

Baksa, Dániel, Aleš Bulíř, and Dyna Heng (2020). *A Simple Macrofiscal Model for Policy Analysis: An Application to Cambodia*. IMF Working Paper WP/20/202. International Monetary Fund.

Banco Central de Timor-Leste (2024). *Monetary Survey: Domestic Credit to the Private Sector*. Online database. Monthly series, end-of-quarter values; accessed 25 October 2026. URL: https://www.bancocentral.tl/.

Blanchard, Olivier and Roberto Perotti (2002). "An empirical characterization of the dynamic effects of changes in government spending and taxes on output". In: *The Quarterly Journal of Economics* 117.4, pp. 1329–1368.

Cholette, Pierre A. (1984). "Adjusting Sub-annual Series to Yearly Benchmarks". In: *Survey Methodology* 10.1, pp. 35–49. URL: https://www150.statcan.gc.ca/n1/pub/12-001-x/1984001/article/14348-eng.pdf.

Denton, Frank T. (1971). "Adjustment of Monthly or Quarterly Series to Annual Totals: An Approach Based on Quadratic Minimization". In: *Journal of the American Statistical Association* 66.333, pp. 99–102. DOI: 10.1080/01621459.1971.10482227.

Dickey, David A. and Wayne A. Fuller (1979). "Distribution of the Estimators for Autoregressive Time Series with a Unit Root". In: *Journal of the American Statistical Association* 74.366, pp. 427–431. DOI: 10.1080/01621459.1979.10482531.

Dime, Roselle, Edimon Ginting, and Juzhong Zhuang (2021). *Estimating Fiscal Multipliers in Selected Asian Economies*. ADB Economics Working Paper Series 638. Asian Development Bank. DOI: 10.22617/WPS210309-2.

Direcção Nacional de Estatística (2025). *Consumer Price Index (CPI Publish Readj New Weight)*. Data file. File `CPI Publish Readj New Weight.xls`, sheet "More"; source indicated in the file: Direcção Nacional de Estatística.

Frankel, Jeffrey A (2012). "The natural resource curse: A survey of diagnoses and some prescriptions". In: *Commodity Price Volatility and Inclusive Growth in Low-Income Countries*, pp. 7–34.

Giannone, Domenico, Michele Lenza, and Giorgio E. Primiceri (2015). "Prior Selection for Vector Autoregressions". In: *The Review of Economics and Statistics* 97.2, pp. 436–451. DOI: 10.1162/REST_a_00483.

Ilzetzki, Ethan, Enrique G Mendoza, and Carlos A Végh (2013). "How big (small?) are fiscal multipliers?" In: *Journal of Monetary Economics* 60.2, pp. 239–254.

Instituto Nacional de Estatística (2024). *National Accounts 2004–2023*. Statistical release, Table 7.1. Quarterly GDP by expenditure at constant 2015 prices; accessed 25 October 2026. URL: https://www.laohamutuk.org/DVD/DGS/NatlAccts2023en.pdf.

International Monetary Fund (2024). *Timor-Leste: Selected Issues*. Country Report 24/343. International Monetary Fund. URL: https://www.imf.org/-/media/files/publications/cr/2024/english/1tlsea2024003-print-pdf.pdf.

International Trade Centre (2025). *Trade Map: Trade Statistics for International Business Development*. Online database. Source file: `timor-leste-imports-exports-to-timor-leste-by-product_all.csv`; accessed 20 October 2026. URL: https://www.trademap.org.

Jordà, Òscar (2005). "Estimation and Inference of Impulse Responses by Local Projections". In: *American Economic Review* 95.1, pp. 161–182. DOI: 10.1257/0002828053828518.

Kuschnig, Nikolas and Lukas Vashold (2021). "BVAR: Bayesian Vector Autoregressions with Hierarchical Prior Selection in R". In: *Journal of Statistical Software* 100.14, pp. 1–32. DOI: 10.18637/jss.v100.i14.

Kwiatkowski, Denis, Peter C. B. Phillips, Peter Schmidt, and Yongcheol Shin (1992). "Testing the Null Hypothesis of Stationarity against the Alternative of a Unit Root". In: *Journal of Econometrics* 54.1–3, pp. 159–178. DOI: 10.1016/0304-4076(92)90104-Y.

Litterman, Robert B. (1986). "Forecasting with Bayesian Vector Autoregressions—Five Years of Experience". In: *Journal of Business and Economic Statistics* 4.1, pp. 25–38. DOI: 10.1080/07350015.1986.10509491.

Mountford, Andrew and Harald Uhlig (2009). "What are the effects of fiscal policy shocks?" In: *Journal of Applied Econometrics* 24.6, pp. 960–992.

Sax, Christoph and Peter Steiner (2013). "Temporal Disaggregation of Time Series". In: *The R Journal* 5.2, pp. 80–87. DOI: 10.32614/RJ-2013-028.

— (2025). *tempdisagg: Methods for Temporal Disaggregation and Interpolation of Time Series*. R package version 1.2.0. URL: https://CRAN.R-project.org/package=tempdisagg.

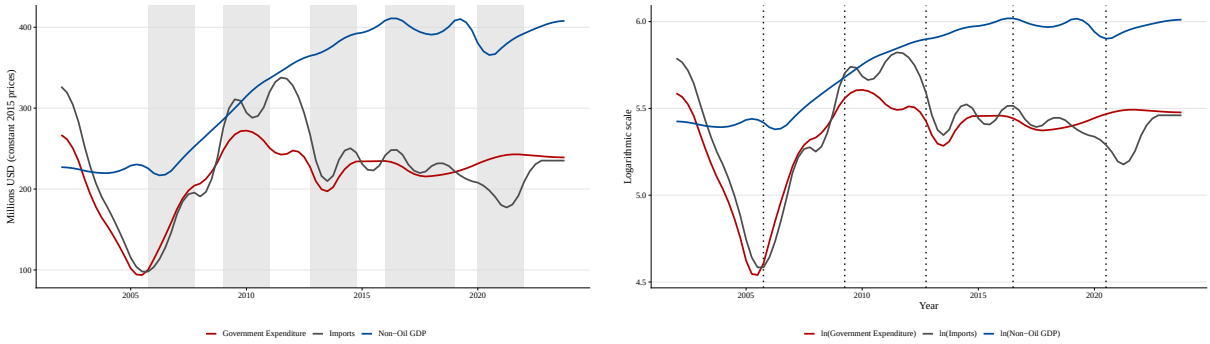
Tang, Hsiao Chink, Philip Liu, and Eddie C. Cheung (2013). "Changing Impact of Fiscal Policy on Selected ASEAN Countries". In: *Journal of Asian Economics* 24, pp. 103–116. DOI: 10.1016/j.asieco.2012.07.003.

World Bank (2021). *Timor-Leste Public Expenditure Review: Changing Course Towards Better and More Sustainable Spending*. Public Expenditure Review. World Bank. URL: https://documents1.worldbank.org/curated/en/619521636038149163/pdf/Timor-Leste-Public-Expenditure-Review-Changing-Course-Towards-Better-and-More-Sustainable-Spending.pdf.

Zivot, Eric and Donald W. K. Andrews (1992). "Further Evidence on the Great Crash, the Oil-Price Shock, and the Unit-Root Hypothesis". In: *Journal of Business and Economic Statistics* 10.3, pp. 251–270. DOI: 10.1080/07350015.1992.10509904.

Appendix A Long-sample descriptive evidence

Figure 6 shows the long-sample series in levels and logs. The series are smooth by construction. The dotted lines in panel (b) mark the main breaks selected by BIC across the five series using the Bai–Perron procedure (Bai and Perron 1998; Bai and Perron 2003). Table 12 reports the unit-root tests for the long sample. The evidence is again mixed, and the BVAR is estimated in levels with a Minnesota prior.



(a) Levels **(b)** Logarithms and breaks

Figure 6: Long-sample macroeconomic series, 2002Q1–2023Q4. Gray areas mark event windows; dotted lines mark the main breaks selected by BIC across the five series. Source: Author's elaboration.

Appendix B Additional results

Figure 7 shows the companion roots for the official three-variable model. All roots lie inside the unit circle. Figure 8 compares the FEVDs of the official and long three-variable models. Figures 9 and 10 present the complete IRF matrices for the official and long three-variable models.

Table 12: Unit-root and stationarity tests, long sample

Series	ADF (p)	KPSS (p)	ZA (t)	5% CV	Break	Stationary?
`ln_gov_exp`	0.085	0.016	−5.09	−5.08	2005Q1	Yes
`ln_gdp_non`	0.943	0.010	−3.37	−5.08	2007Q2	No
`ln_imports`	0.381	0.100	−3.68	−5.08	2008Q1	No
`ln_cpi`	0.599	0.010	−4.02	−5.08	2009Q4	No
`ln_credit`	0.010	0.010	−12.66	−5.08	2003Q1	Yes

Note: See the notes to Table 6. Source: Author's elaboration.

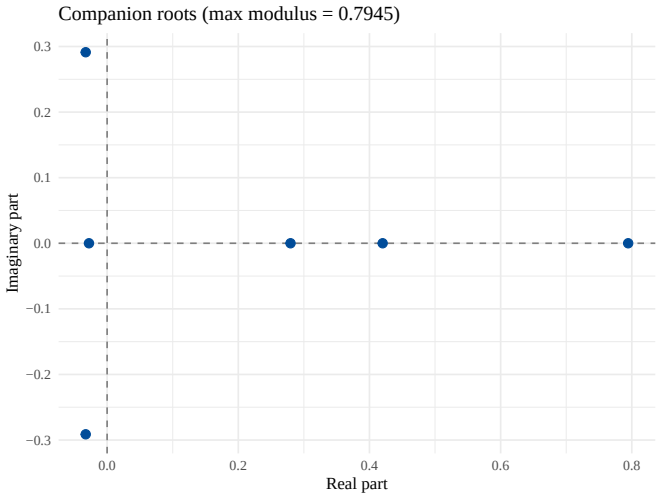
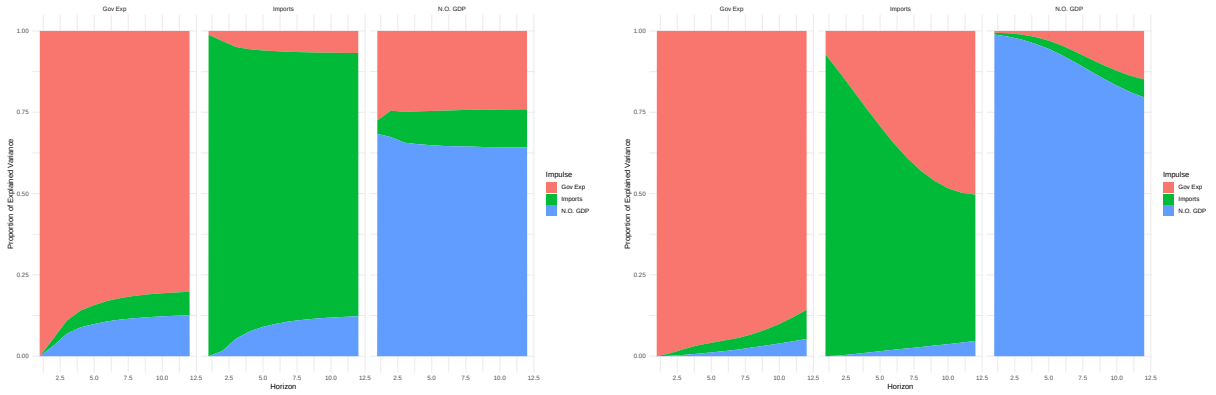


Figure 7: Companion roots of the official three-variable BVAR. The maximum modulus is 0.795, below one. Source: Author's elaboration.



(a) Official sample **(b)** Long sample

Figure 8: Forecast-error variance decomposition, three-variable BVAR, order 1. Source: Author's elaboration.

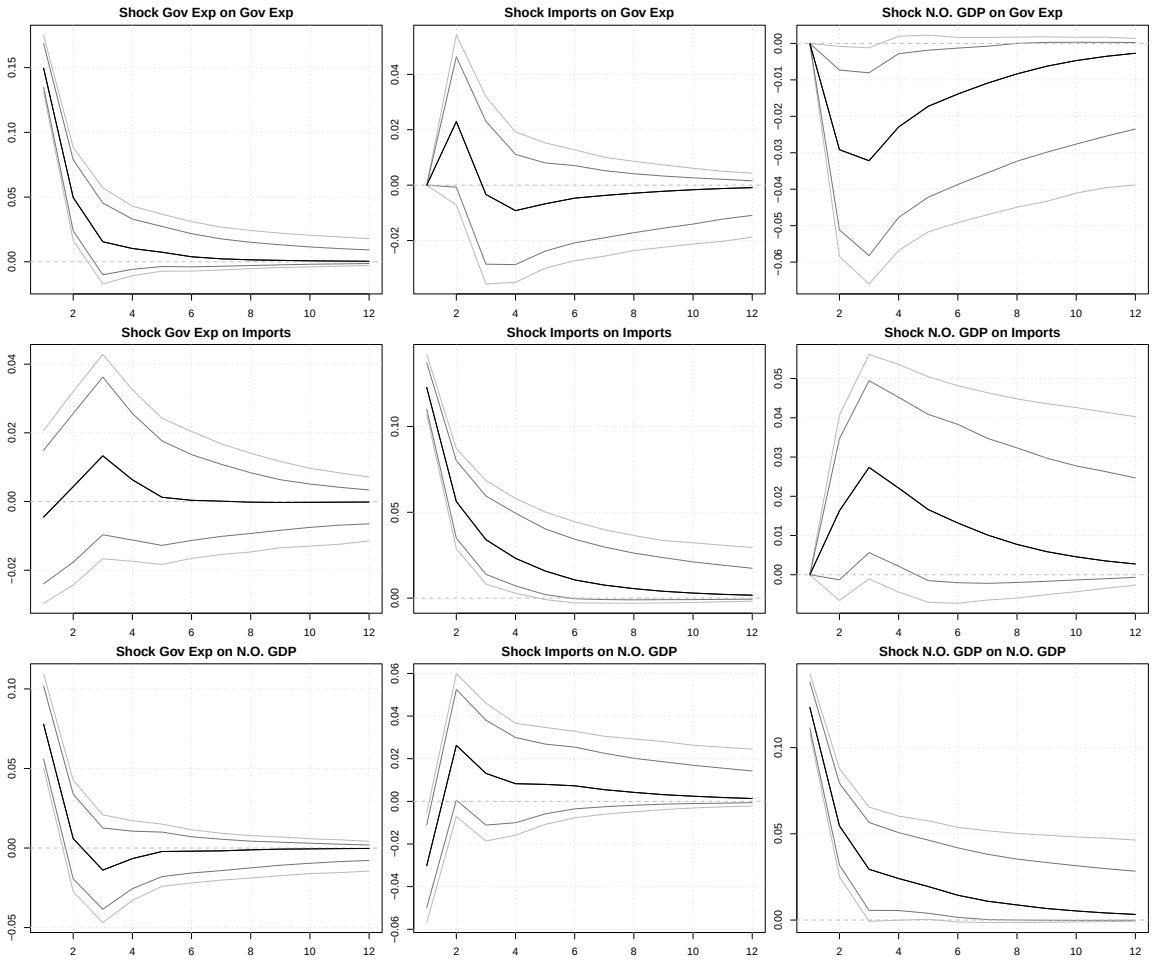


Figure 9: Complete matrix of orthogonalized IRFs, official sample, order 1. Each panel shows the response of the variable after "on" to a shock to the variable after "Shock". Bands: 68% and 90%. Source: Author's elaboration.

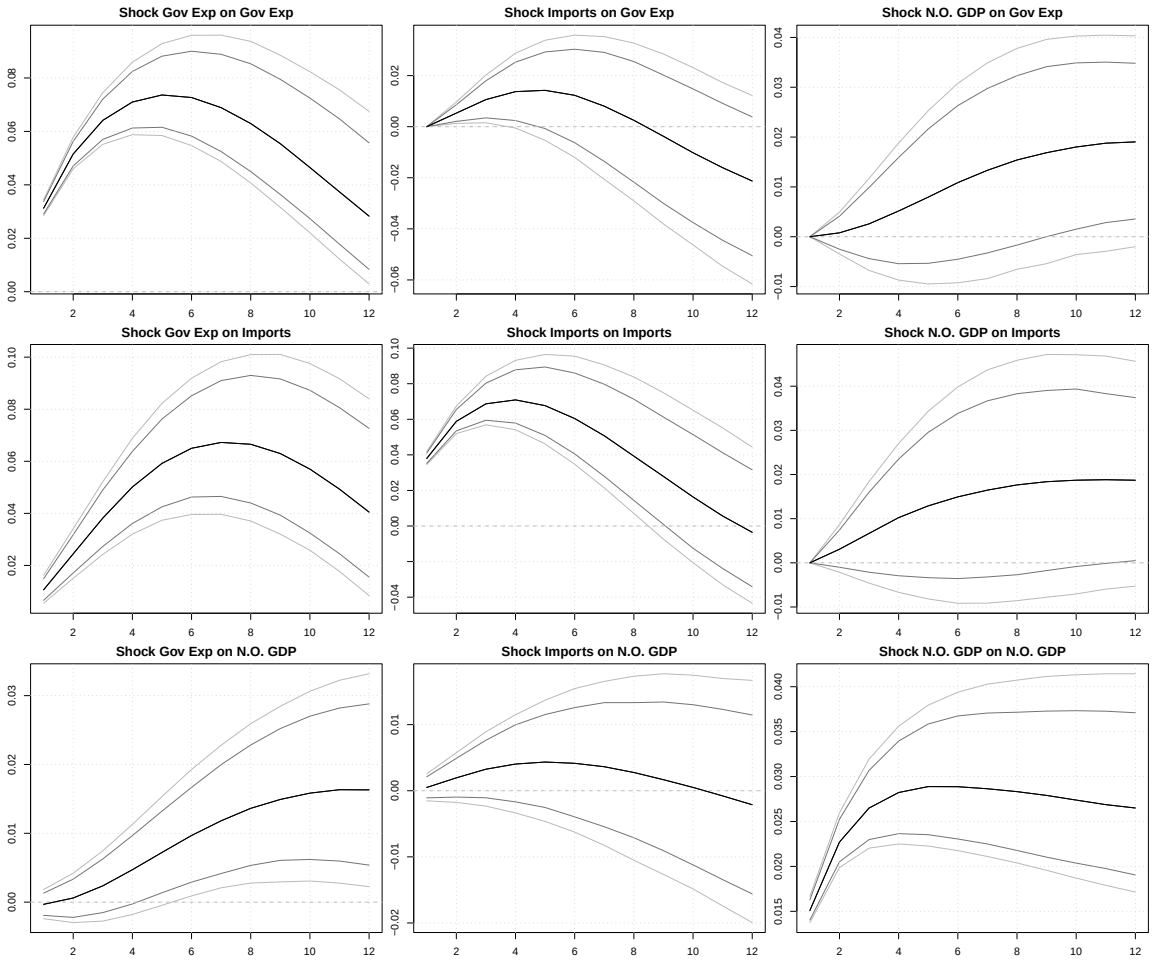


Figure 10: Complete matrix of orthogonalized IRFs, long sample, order 1. Each panel shows the response of the variable after "on" to a shock to the variable after "Shock". Bands: 68% and 90%. Source: Author's elaboration.